Linguistics / Cognitive Science c142

Week of Tue Aug 29:
Language and animal communication

Terry Regier
GSI: Noah Hermalin

Bureaucracy

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- **Discussion check-ins**: On both Tuesday and Thursday, there will be short online discussion check-ins.
- **Miss-able assignments**: 1 quiz, + 2 check-ins, no questions asked.

Rest assured

- Class sessions are **not** being recorded.
- “Rest assured that no recordings will take place unless you sign up” [for Course Capture, for which I did **not** sign up].

This week’s readings

- Hockett: Design features of language
- NYT: Chimp talk debate
- Savage-Rumbaugh et al.: Kanzi etc.

The plan for this week

- Warm-up for quiz, small groups
- Quiz
- Background mini-lecture
- Discussion of this week’s readings
- Integrative discussion

Small groups

Ask each other questions about the readings, in preparation for the quiz.

Quiz on bCourses

Open notes.
Your time limit is shown on your quiz. Reconvene in 10 mins.

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Please start **now**.

Background mini-lecture

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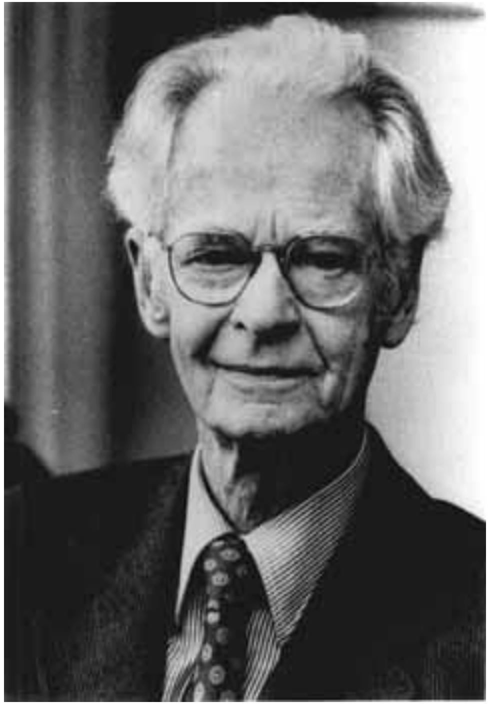
Origins of human knowledge

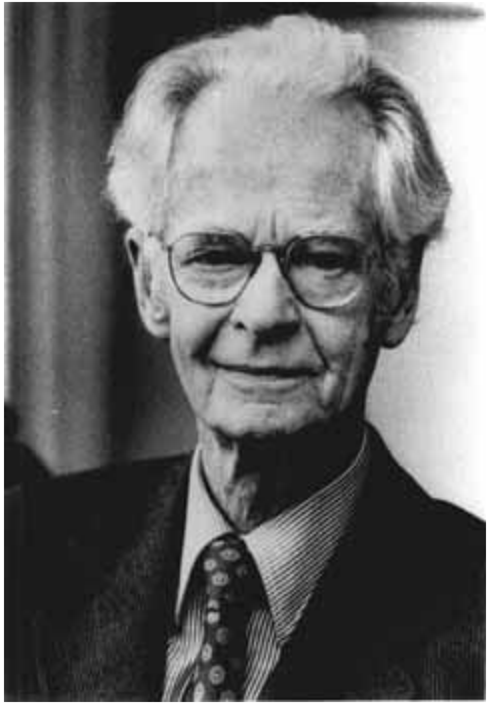
- **Nativism**: Some forms of knowledge are innate, and may be species-specific.
- **Empiricism**: All knowledge is based in experience.
- This opposition (see here for more detail) will be relevant throughout our course.

A prominent empiricist



B.F. Skinner
(1904-1990)

A prominent empiricist



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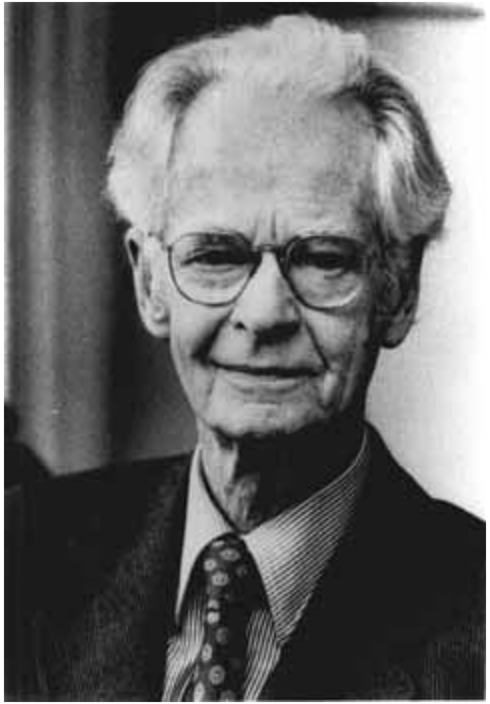

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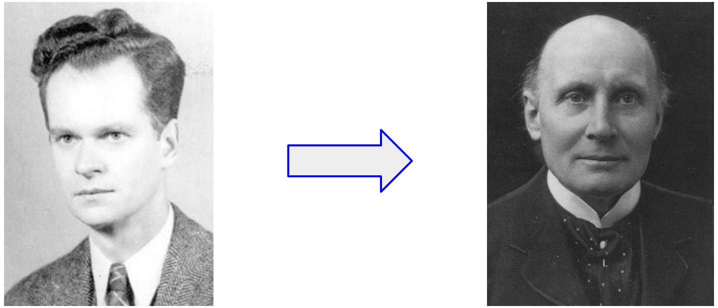
- Eminent intellectual, older and more well-known than Skinner.
- Seated **next to Skinner** at a dinner table in 1934.

The dinner conversation, 1934, as recalled by Skinner

We dropped into a discussion of behaviorism … of which I was a zealous devotee … I began to set forth the principal arguments of behaviorism with enthusiasm.



The dinner conversation, 1934, as recalled by Skinner

Professor Whitehead was equally in earnest. ... He agreed that **science** might be successful in accounting for human behavior provided one made an **exception** …

Verbal behavior

[With respect to **verbal** behavior], he insisted, something else must be at work. He brought the discussion to a close with a friendly challenge…

Whitehead’s challenge

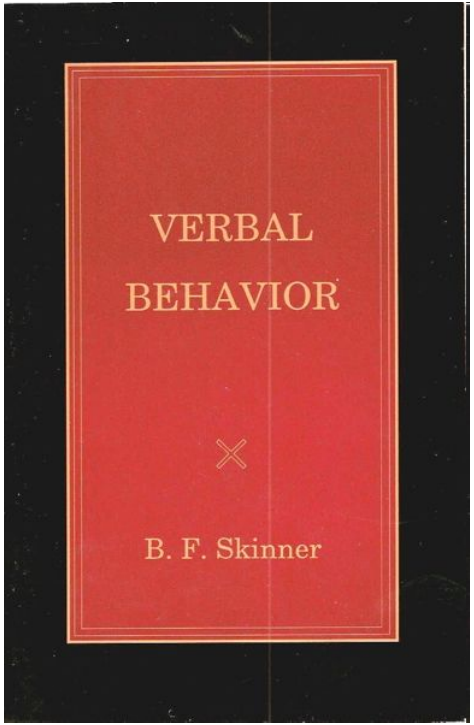
“Let me see you,” he said, “account for my behavior as I sit here saying **‘No black scorpion is falling upon this table.’**”

Whitehead’s challenge

Captures the *mental state*
encoding the *absence*
of a scorpion.

Skinner’s response

- Published 1957: **23 years** after dinner with Whitehead!



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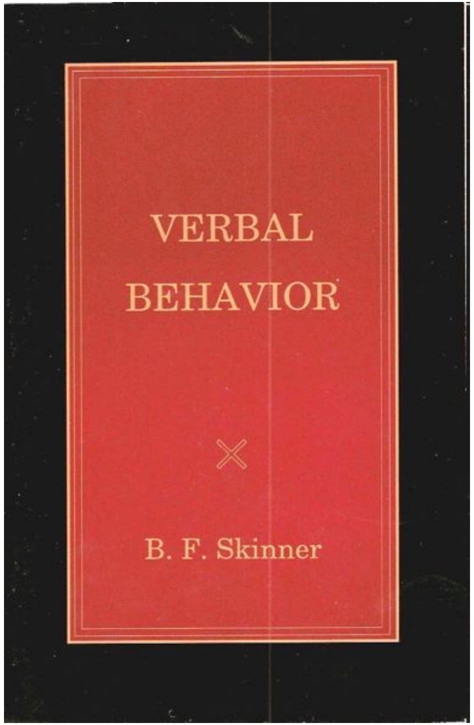
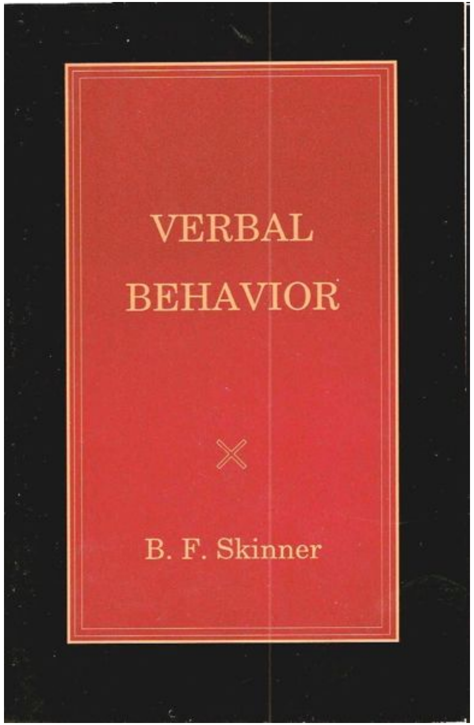
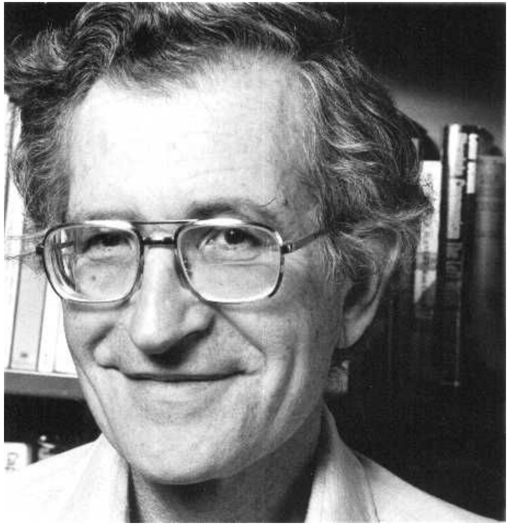


Skinner’s response

- Published 1957: **23 years** after dinner with Whitehead!
- Proposes an account of human language in behaviorist terms.
- **Negatively reviewed** by Noam Chomsky in 1959.





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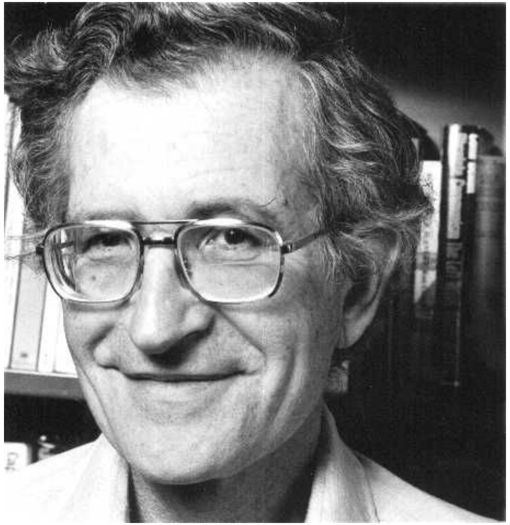

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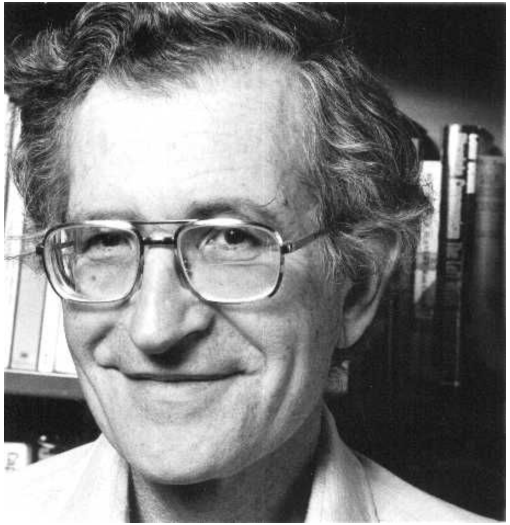

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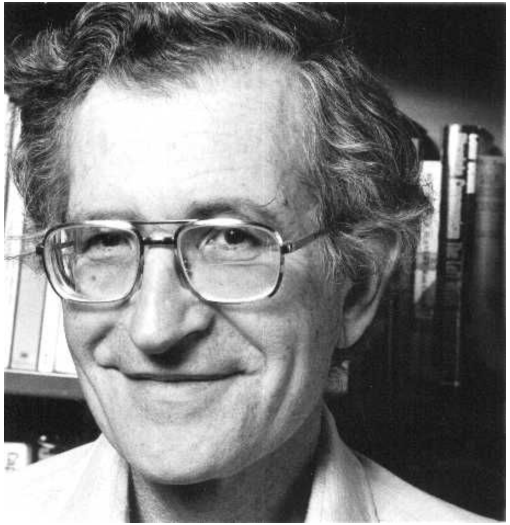

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- Associations are **inadequate**.

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- But the **attack on associationism** is relevant to understanding the broader significance of this week’s readings.

Discussion of this week’s readings

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$\boxed{/\ \text{d ɑː g}\ /}$

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Phonemes: meaningless but meaning-differentiating ingredients

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Duality of patterning

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6. Displacement: removed in time and space.
7. Cultural transmission: rather than genetically determined.

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Highlights

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- Moving the goal posts?
- Earlier efforts with other species.
- Keyboard instead of manual signing.
- Desire to knock people off their self-appointed thrones.

Discuss in small groups

According to Chomsky (as quoted in the NYT piece): “attempting to teach linguistic skills to animals is irrational – like trying to teach people to flap their arms and fly. Humans can fly about 30 feet – that's what they do in the Olympics. Is that flying? The question is totally meaningless.”

Do you agree? Disagree? On the fence? **Why?**

Welcome to Thursday!

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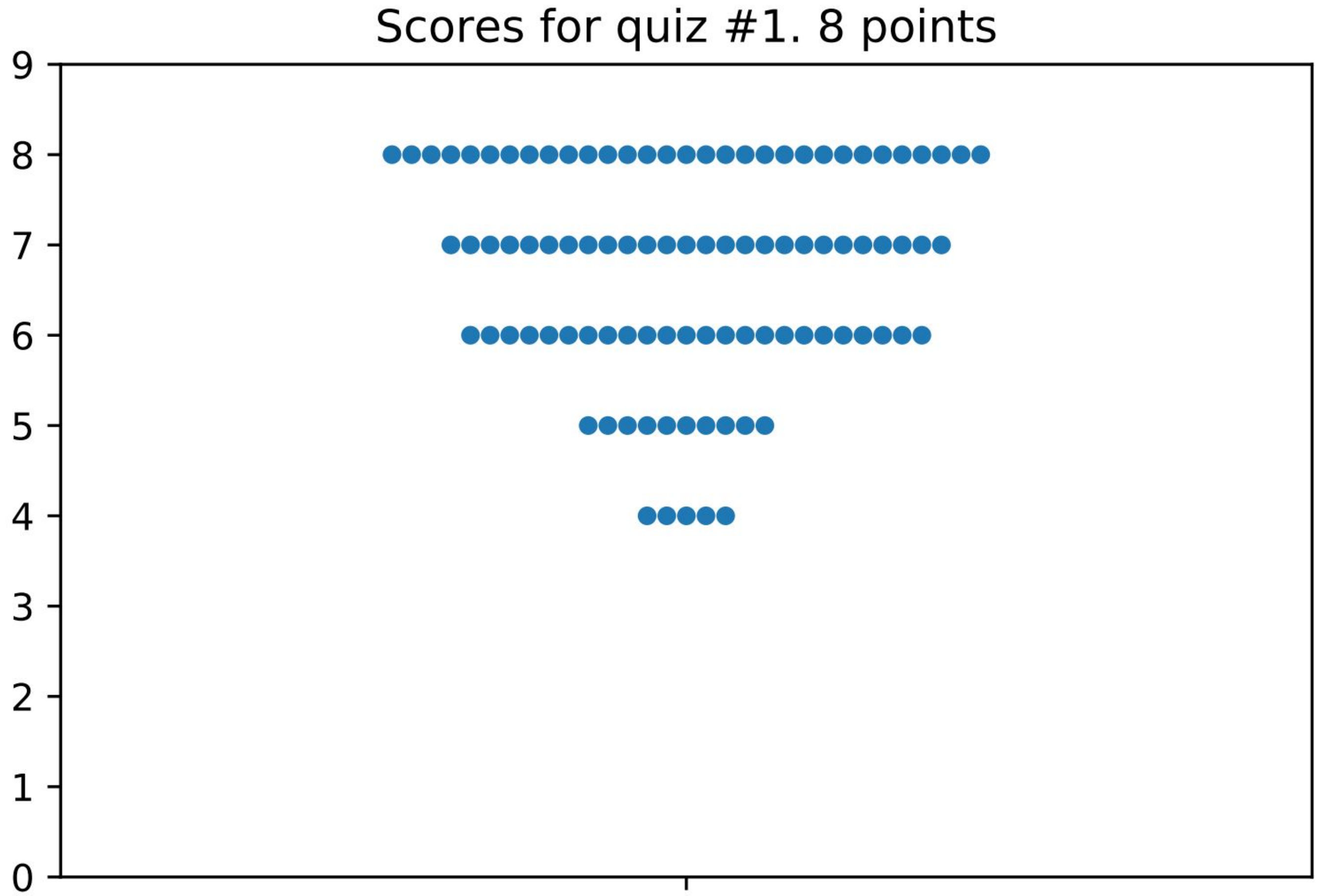
Scores for quiz #1. 8 points
9
8
7
6
5
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3
2
1
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A useful attitude

About **mistakes**, and making the best of it.

Reactions?

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STUART CONWAY

Savage-Rumbaugh et al.

- Background
- Hypotheses and predictions
- Methods
- Results
- Conclusions and theoretical contributions

Savage-Rumbaugh et al.

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I won’t always give you this much scaffolding!

Background

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There is a common view that:

- Very young children use words to **request** things – e.g. “milk” might mean “give me milk” – as if they had simply **associated** the word with the desired action.

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- Very young children use words to **request** things – e.g. “milk” might mean “give me milk” – as if they had simply **associated** the word with the desired action.
- In contrast, older children use words **referentially**, i.e. to name things without requesting them.

Background

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- **Non-human** primates had previously been taught to produce symbols in order to request things, but this had required **explicit training**.
- And once trained, non-human primates had been shown to engage in **referential symbol use**.
- But there had been **no** demonstrations of non-human primates **spontaneously** learning a symbol system without explicit training, and spontaneously using it **referentially**.

Hypotheses and predictions

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?

Methods

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- Kanzi is a **bonobo**, or pygmy chimpanzee.

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- He was reared in an environment in which his **mother** was being trained to use symbols (**lexigrams**).

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- He was reared in an environment in which his **mother** was being trained to use symbols (**lexigrams**).
- His grasp of lexigrams and of English words was tested **formally** (by either presenting him with a word or symbol and asking him to indicate the corresponding object, or vice versa), and **informally**, by tracking his use of symbols in day-to-day interactions.

Methods

- His younger sister Mulika was similarly tested.
- Synthesized as well as natural speech was used.

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- His **mother's performance** was not comparable to his, suggesting that there is a **sensitive period** for symbol learning in young bonobos.

Results

- Kanzi originally learned words e.g. “strawberry” in the context of **routines** such as going to where there are strawberries: a “complexive” or **associative** meaning.

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- Kanzi originally learned words e.g. “strawberry” in the context of **routines** such as going to where there are strawberries: a “complexive” or **associative** meaning.
- He only later learned to **break out** of these routines, understanding e.g. “hide strawberry”, in which the symbol for “strawberry” **does not request** the strawberry itself, but rather **indicates** it so that some other action may be performed on it.

Results

- This latter stage is understood as **referential**, de-complexified, and allows multi-symbol utterances such as “**no bite mushrooms**”, in which Kanzi shows that he understands that he is not supposed to eat wild mushrooms outdoors.

Captures the *mental state*
encoding the *absence*
of a scorpion.

Results

- This latter stage is understood as **referential**, de-complexified, and allows multi-symbol utterances such as “**no bite mushrooms**”, in which Kanzi shows that he understands that he is not supposed to eat wild mushrooms outdoors.
- Kanzi also produced some multi-symbol utterances, e.g. “**chase person1 person2**”, in which he was neither person. Such utterances were **his inventions**, not imitations.

Conclusions

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There are several important **differences** between Kanzi’s symbolic abilities and those of common chimpanzees.

Conclusions

1. Kanzi easily understood that symbols can refer to absent referents and events (displacement).
2. Kanzi did not need to be taught the difference between naming and requesting.
3. Kanzi understood English words, and has fine-grained semantic discrimination – e.g. he does not confuse coke with juice; he is not semantically sloppy.
4. Kanzi requested that A act on B, when he was neither A nor B.

Conclusions

Thus, bonobos, as exemplified by Kanzi, seem to have some limited ability to use symbols to communicate – more so than other non-human primates that have been tested.

Savage-Rumbaugh et al.

- Background
- Hypotheses and predictions
- Methods
- Results
- Conclusions and theoretical contributions

Some pushback!

- Susan Goldin-Meadow: Only **4%** of Kanzi’s utterances are comments on the state of the world around him – most are requests, demands, etc. **He doesn’t want to chat** – unlike human children.

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- Susan Goldin-Meadow: Only **4%** of Kanzi's utterances are comments on the state of the world around him – most are requests, demands, etc. **He doesn't want to chat** – unlike human children.
- Why don't bonobos use symbolic communication among themselves, in the wild?

Discussion check-in coming up

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We are about to break into small groups to discuss something, followed by yes / no / on-the-fence and full-class discussion, as before.

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We are about to break into small groups to discuss something, followed by yes / no / on-the-fence and full-class discussion, as before.

Discussion check-in happens **after that**: please record a point that occurred to you, and a point that occurred to someone else in your discussion group. We'll show you how.

Integrative discussion: small groups

Cast your mind back over **everything** we have discussed this week – empiricism vs. nativism, Skinner vs. Whitehead, Hockett, design features, Chomsky, universal grammar, Savage-Rumbaugh, associations vs. reference, Kanzi's abilities and non-abilities – and pull out of it what you take to be a **major and important take-home message**, and say why it is important.

Discussion check-in

bCourses > Assignments > Discussion

Select: “Discussion check-in: August 31”

We expect this to take around 2 minutes - if it takes a bit longer that's OK - look up at me when you're done.

Please record a point that occurred to you, and a point that occurred to someone else in your discussion group.

Major points of this week

- Nativism vs. empiricism
- Skinner, Whitehead, Chomsky
- Hockett and design features of language
- Duality of patterning and large lexicons
- Displacement, gossip, business, marriage
- Associations vs. reference
- Kanzi’s abilities and inabilities

Week of Tue Sep 5: The faculty of language

- Chomsky, N. (1986). Preface & Knowledge of language as a focus of inquiry.
- Hauser, M. D. et al. (2002). The faculty of language: What is it, who has it, and how did it evolve?
- Pinker, S. & Jackendoff, R. (2005). The faculty of language: What's special about it?